

Quantifying Data Leakage in EEG Emotion Classification: A Four-Tier Evaluation Framework

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Abstract—EEG-based emotion recognition studies frequently report classification accuracies exceeding 90% on benchmark datasets, yet these results often rely on evaluation protocols that violate sample independence assumptions. We propose a four-tier evaluation framework that systematically quantifies how evaluation methodology inflates reported performance. Using the DEAP dataset (20 subjects, 32 EEG channels) and a custom OpenBCI dataset (1 subject, 16 channels), both employing music-evoked emotion stimuli mapped to Russell’s arousal-valence circumplex—we evaluate an MLP baseline and a Graph Attention Network (GAT) under identical conditions for both 4-class quadrant and binary (arousal/valence) classification. Evaluation protocol selection dominates reported performance: 4-class macro F1 degrades from 0.607 (Tier 0, random split) to 0.311 (Tier 2, trial-aware) to 0.210 (Tier 3, LOSO), a 39.7-point drop. Binary valence classification yields F1=0.649 under trial-aware evaluation, enabling direct comparison with published benchmarks. Trial leakage ($\Delta = -0.337$, Cohen’s $d = 3.21$, $p < 0.001$) is the dominant inflation mechanism. On the personalized OpenBCI dataset, both models achieve >97% macro F1 under trial-aware evaluation, demonstrating that EEG-BCI may be viable when inter-individual variability is eliminated.

Index Terms—EEG, emotion recognition, graph attention networks, evaluation methodology, data leakage, brain-computer interface.

I. INTRODUCTION

EEG-based emotion recognition has attracted considerable research attention, with numerous studies reporting classification accuracies between 85–99% on benchmark datasets such as DEAP [1]. However, many results are obtained using evaluation protocols that inadvertently violate the independence assumption between training and test samples, leading to severely inflated performance estimates [2], [3]. Recent analyses corroborate this: Zhang *et al.* [4] observe that high accuracy claims often overshadow methodological limitations, Khan *et al.* [5] found that 87% of 101 reviewed DEAP studies contained methodological errors—including data leakage and biased feature selection—inflating accuracy by up to 46%, and Lei *et al.* [6] provided direct empirical evidence that trial-wise leakage inflates EEG emotion classification performance.

The core issue is *trial leakage*: when EEG signals are segmented into overlapping windows and these windows are randomly assigned to training and test sets without respecting trial boundaries, near-duplicate feature vectors appear in both sets. A classifier can exploit temporal autocorrelation rather

than learning genuine emotional representations. Even moderate overlap (50%, as used in our study) produces correlated feature vectors across adjacent windows; extreme overlap used in some studies [7] exacerbates this problem dramatically.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the four-tier methodology, feature extraction and datasets. Section IV presents experimental results for both 4-class and binary classification. Section V discusses implications. Section VI concludes.

This paper makes three contributions:

- 1) A **four-tier evaluation framework** that decomposes EEG emotion classification into progressively rigorous protocols, enabling precise quantification of leakage inflation.
- 2) **Empirical evidence** that evaluation protocol selection dominates reported performance – more so than any architectural improvement.
- 3) **Architecture comparison**: under trial-aware evaluation, both MLP and GAT are evaluated with matched feature sets for 4-class and binary tasks across two datasets sharing the same music-evoked emotion paradigm.

II. RELATED WORK

A. EEG Emotion Recognition

The DEAP dataset [1] is the most widely used benchmark for EEG emotion recognition, featuring 32 subjects watching 40 one-minute music videos while 32-channel EEG is recorded. Self-reported arousal and valence ratings (1–9 scale) serve as ground truth. Published results vary widely: the original paper reports 57.5–62.0% binary accuracy under subject-dependent evaluation, while subsequent deep learning studies report 90–99% [7]. This discrepancy between rigorous and random-split evaluation represents the critical confound our framework quantifies. Zhong *et al.* [8] applied regularized Graph Neural Networks on SEED, demonstrating graph-based spatial modeling under cross-subject evaluation. Zheng and Lu [9] investigated critical frequency bands for deep neural network-based emotion recognition, also on SEED. Neither reports results on DEAP’s arousal-valence formulation, leaving a gap in directly comparable benchmarks that our framework addresses.

Several works address cross-subject generalization through architectural or normalization innovations. Apicella *et al.* [10] systematically reviewed 75 studies on cross-subject and cross-session generalization, concluding that transfer learning methods are most effective. Fdez *et al.* [11] proposed stratified normalization to reduce inter-participant variability, achieving 91.6% cross-subject accuracy on SEED. Li *et al.* [12] exploited cerebral hemispheric asymmetry through BiDANN, Du *et al.* [13] combined attention-based LSTM with domain discrimination, and Ning *et al.* [14] applied domain adaptive few-shot learning for cross-subject transfer. Kukhila *et al.* [15] proposed EEGain, a unified open-source evaluation framework standardizing preprocessing, data splitting, and metrics across six benchmark datasets—independently motivating the same evaluation standardization goals as our four-tier framework.

B. Evaluation Methodology

Data leakage in neuroimaging classification is well-documented. Varoquaux *et al.* [2] demonstrated that improper cross-validation inflates brain decoding accuracy by exploiting temporal structure in the data. Roberts *et al.* [3] showed that temporal and hierarchical data structures require specialized CV strategies that respect the dependency structure of observations. Our four-tier framework instantiates these principles specifically for EEG-BCI, extending prior work by separately quantifying trial leakage versus cross-subject generalization difficulty.

C. Graph Neural Networks for EEG

Graph Attention Networks (GATs) [16] naturally model EEG channel relationships by treating electrodes as graph nodes. Each GAT layer computes attention coefficients between channel pairs, learning which inter-electrode relationships are most informative for classification. Recent GNN applications to EEG emotion recognition show promising within-subject results [8], but limited investigation under rigorous cross-trial or cross-subject evaluation. We employ a 3-layer multi-head GAT with fully-connected adjacency to test whether attention-based spatial modeling can overcome the generalization challenge, and compare against a parameter-matched MLP to isolate the contribution of graph structure.

III. METHODOLOGY

A. Four-Tier Evaluation Protocol

We define four tiers of increasing methodological rigor. The temporal overlap between adjacent windows is:

$$\text{Overlap}(S) = \frac{W - S}{W} \quad (1)$$

where W is the window length (2 s) and S is the step size. With $S = W/2$ (our configuration), overlap is 50%. A random 80/20 split places adjacent windows in different sets with probability $2 \times 0.8 \times 0.2 = 0.32$, enabling exploitation of shared signal content.

Tier 0 (Maximum Leakage): All subjects pooled. Windows extracted with 50% overlap. Random 80/20 split without respecting subject or trial boundaries.

TABLE I
MODEL ARCHITECTURES

Component	Configuration	Params
<i>DeepGAT (fully-connected graph)</i>		
GAT Layers	3 layers, 4 heads, dim=96	~126K
Graph	Fully-connected ($C \times C$)	—
Classifier	Dense($96 \rightarrow n_c$)	included
<i>MLP Baseline</i>		
Hidden Layers	$C \times 26 \rightarrow 128 \rightarrow 64 \rightarrow n_c$	~116K

Tier 1 (Within-Subject, Leaky): Per-subject 10-fold StratifiedKFold with 50% overlap. Trial boundaries not enforced, so windows from the same trial may appear in different folds.

Tier 2 (Trial-Aware): Per-subject StratifiedGroupKFold where groups = trials. No temporal leakage: all windows from a given trial are assigned to the same fold. This is the minimum standard for valid within-subject evaluation.

Tier 3 (Cross-Subject LOSO): Leave-One-Subject-Out. Trains on 19 subjects, tests on the held-out subject. Evaluates cross-subject generalization.

B. DeepGAT Architecture

The DeepGAT model treats EEG channels as nodes in a fully-connected graph. Table I summarizes both architectures.

Hyperparameters were optimized using Optuna [17] (TPE sampler, 150 trials) with a combined objective: $(F1_{T1} + F1_{T2} + F1_{T3})/3$. Best configuration: $\text{lr}=3.9 \times 10^{-3}$, $\text{batch}=128$, $\text{label smoothing}=0.105$, $\text{input noise } \sigma=0.13$, $\text{attention dropout}=0.197$.

C. Feature Extraction

We extract 26 features per channel:

- **Spectral:** Band Power (5) and Differential Entropy (5) across $\theta(4-8)$, $\alpha(8-12)$, $\beta_l(12-16)$, $\beta_h(16-25)$, $\gamma(25-45 \text{ Hz})$
- **Connectivity:** Phase-Locking Value [18] (5) and Coherence (5), computed as mean per-channel connectivity
- **Cross-frequency:** Phase-Amplitude Coupling [19] (θ - γ , 1 feature)
- **Temporal:** Sub-window band power variability (5)

Total feature dimensionality: $C \times 26$ (832 for DEAP, 416 for OpenBCI). Window length is 2 s with 50% overlap (step = 1 s) for both datasets.

D. Datasets

Both datasets employ music as the emotion-induction stimulus, mapping to Russell’s circumplex model of affect [20]. This shared paradigm enables controlled comparison of cross-subject versus personalized classification while holding the stimulus modality constant.

The DEAP subjects (s01–s20) were selected sequentially as a 20-subject subset due to computational constraints; the full DEAP corpus contains 32 subjects. Labels derive from self-report arousal and valence ratings using per-subject median splits, forming four quadrants: HAHV, HALV, LAHV, LALV.

TABLE II
DATASET COMPARISON

	DEAP	OpenBCI (Custom)
Subjects	20 (of 32)	1
Channels	32 (BioSemi)	16 (Cyton+Daisy)
Sampling rate	128 Hz	125 Hz
Classes	4 (AV quadrant)	4 (AV quadrant)
Trials	40/subject	100 (25/class)
Task	Music video (40 clips)	Music audio (100 tracks)*
Labels	Self-report (median split)	Experimenter-assigned

*Full stimulus list available at: https://docs.google.com/spreadsheets/d/1xvwLRa_QE4QsopQWWJBOMo-lzheScZMt5OZLdEFLDng

TABLE III
4-CLASS RESULTS (DEAP, 32CH, 26 FEAT/CH, CHANCE = 0.25)

Tier	MLP (~116K)	GAT (~126K)
0 (Max leakage)	0.599 ± 0.010	0.607 ± 0.008
1 (Within-subj., leaky)	0.684 ± 0.103	0.648 ± 0.124
2 (Trial-aware)	0.298 ± 0.054	0.311 ± 0.050
3 (LOSO)	0.200 ± 0.070	0.210 ± 0.060

Music stimuli for the custom OpenBCI dataset were manually selected to target the four quadrants of Russell’s arousal-valence space (calm, happy, sad, stressed), with 25 tracks per category assigned by the experimenter. The single participant was the lead author, who provided informed self-consent. Both approaches map to the same arousal-valence space, with chance level at 25% (4-class) or 50% (binary).

E. Training Protocol

Both models use AdamW with cosine annealing and class-weighted cross-entropy loss. Early stopping (patience=20) prevents overfitting. Feature normalization uses z-score standardization fit exclusively on training data per fold, preventing information leakage through feature statistics. For LOSO, the scaler is fit on the $N-1$ training subjects and applied to the held-out subject. The MLP baseline provides a flat-input comparison to validate that any GAT advantage stems from its graph structure rather than capacity differences.

IV. RESULTS

A. 4-Class Quadrant Classification

Table III presents macro F1 across all tiers. The dominant finding is a 29.6-point F1 drop from T0 to T2, with an additional 10.1-point drop to T3.

The T0→T1 increase ($\Delta_{\text{GAT}} = +0.041$) is not contradictory: Tier 0 pools all subjects in a single random split, introducing implicit cross-subject generalization difficulty; Tier 1 evaluates per-subject, removing this component while retaining trial leakage. The T1→T2 transition ($\Delta_{\text{GAT}} = -0.337$) identifies trial leakage as the primary inflation source. The T2→T3 gap ($\Delta = -0.101$) reflects cross-subject generalization difficulty. Paired t -tests with Bonferroni correction confirm all tier transitions are significant: T1 vs. T2: $t = 13.99$, $p < 0.001$, $d = 3.21$; T2 vs. T3: $t = 5.69$, $p < 0.001$, $d = 1.31$.

TABLE IV
BINARY CLASSIFICATION (DEAP, MACRO F1, CHANCE = 0.50)

Dim.	Tier	MLP	GAT
Arousal	T0 (random)	0.712±0.007	0.700±0.012
	T1 (leaky)	0.768±0.066	0.743±0.084
	T2 (trial-aware)	0.583±0.048	0.597±0.043
	T3 (LOSO)	0.459±0.083	0.516±0.070
Valence	T0 (random)	0.737±0.006	0.739±0.008
	T1 (leaky)	0.799±0.065	0.777±0.074
	T2 (trial-aware)	0.643±0.086	0.649±0.077
	T3 (LOSO)	0.513±0.095	0.555±0.090

TABLE V
OPENBCI WITHIN-SUBJECT RESULTS (4-CLASS, CHANCE = 0.25)

Tier	MLP	GAT
0 (Max leakage)	0.998 ± 0.001	0.997 ± 0.002
1 (Within-subj., leaky)	0.999 ± 0.001	0.998 ± 0.002
2 (Trial-aware)	0.972 ± 0.031	0.981 ± 0.017

Under T2, GAT slightly outperforms MLP ($\Delta = +0.013$, $d = 0.74$, $p < 0.05$), suggesting graph-based spatial modelling provides a genuine but modest benefit once leakage is removed.

B. Binary Classification

To contextualize results against the majority of published EEG-BCI literature, Table IV reports binary arousal and valence classification (per-subject median threshold, chance = 50%) across all four tiers.

The same degradation pattern persists: T1→T2 drops of 14.6 (arousal) and 12.8 (valence) points confirm trial leakage inflates binary results similarly to 4-class. Valence is consistently more discriminable than arousal across all tiers, consistent with frontal EEG asymmetry encoding valence more reliably [21], [22]. Notably, at T2, valence (0.649 GAT) outperforms arousal (0.597 GAT), suggesting valence may be more amenable to EEG-based classification.

The GAT advantage over MLP widens under stricter tiers: at T2, GAT leads by 1.4 (arousal) and 0.6 (valence) points; at T3, this gap widens to 5.7 and 4.2 points respectively. This pattern suggests GAT’s spatial attention captures genuine emotional signal rather than exploiting temporal leakage.

C. Personalized BCI (OpenBCI)

On the custom OpenBCI dataset (single subject, 100 trials, 16 channels), we evaluate under T0–T2 (LOSO is not applicable for $n=1$).

The minimal degradation from T1 (0.998) to T2 (0.981) contrasts sharply with the DEAP result (T1: 0.648 → T2: 0.311). This demonstrates that when inter-individual variability is eliminated, EEG-based emotion classification achieves near-perfect performance even under proper trial-aware evaluation. The 1.7-point T1→T2 drop for OpenBCI versus the 33.7-point drop for DEAP quantifies the relative contributions of trial leakage versus inter-subject variability to performance

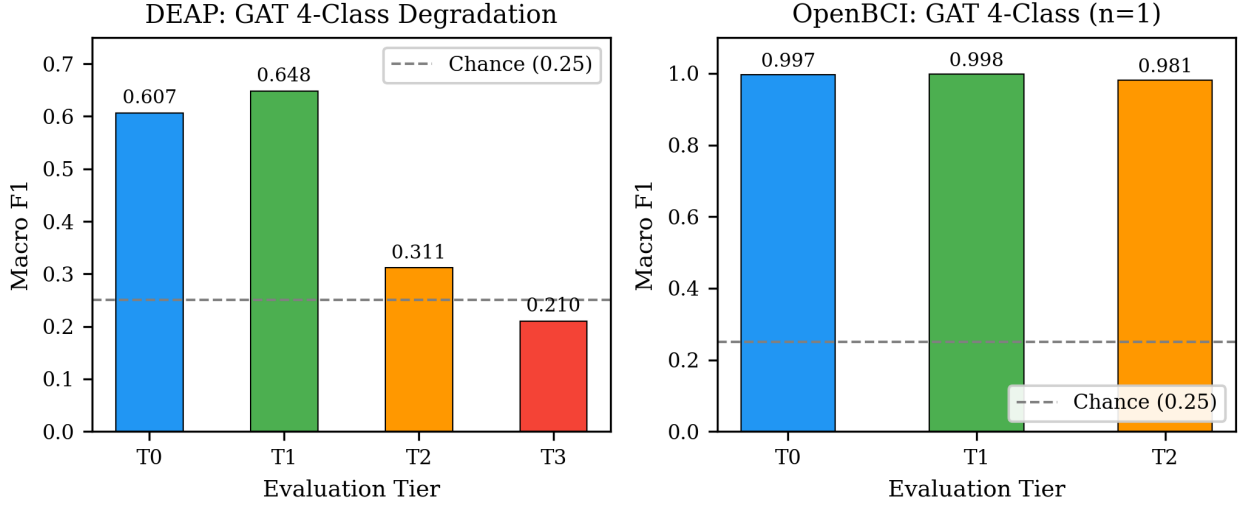


Fig. 1. Progressive performance degradation of the DeepGAT model across evaluation tiers. Left: DEAP 4-class (chance=0.25). The T1→T2 drop identifies trial leakage as the dominant inflation mechanism. Right: OpenBCI personalized 4-class ($n=1$, chance=0.25) maintains >0.98 even under trial-aware evaluation.

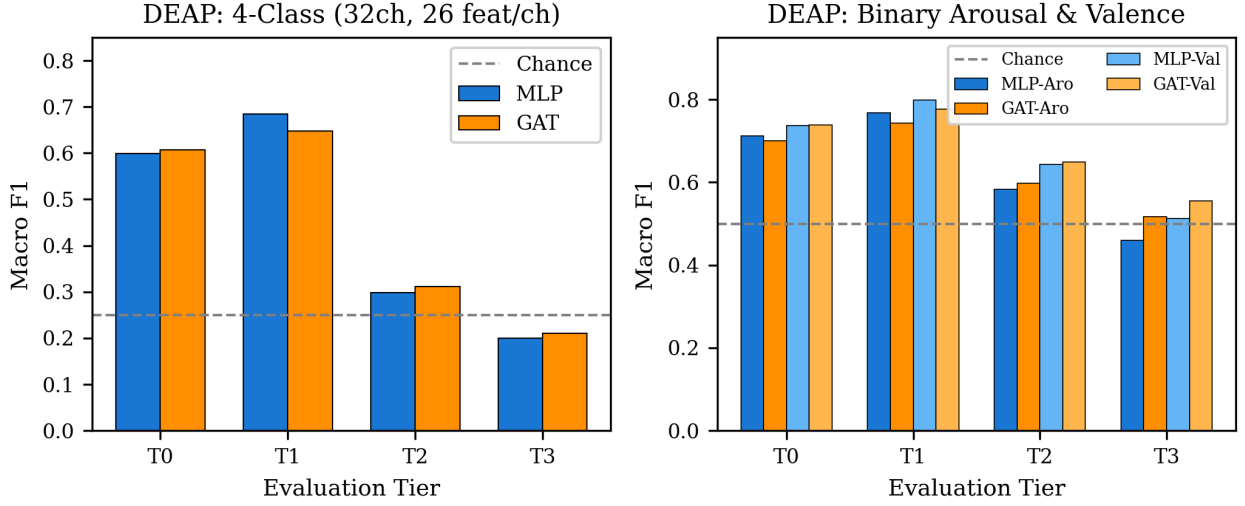


Fig. 2. Model comparison across evaluation tiers. Left: DEAP 4-class (chance=0.25): both models degrade similarly, confirming protocol dominance over architecture. Right: DEAP binary arousal and valence (chance=0.50): valence consistently more discriminable than arousal; GAT advantage emerges under strict evaluation (T2, T3).

inflation. This result ($n=1$) requires multi-subject replication to confirm generalizability.

D. Explainability Analysis

Attention weight analysis of the trained GAT reveals the top-5 most attended channels: CP2, T7, AF4, P3, CP1. The prominence of temporal (T7) and parietal (CP1, CP2, P3) electrodes is consistent with established neuroscience findings linking these regions to emotional processing [23], [24]. Frontal attention asymmetry (AF4) aligns with the valence lateralization hypothesis, which posits that the right prefrontal cortex is more involved in withdrawal-related (negative valence) emotions [21], [22]. The learned attention patterns provide interpretable evidence that the GAT captures physio-

logically meaningful spatial relationships rather than arbitrary feature correlations.

V. DISCUSSION

A. Protocol Dominance Over Architecture

Evaluation methodology accounts for a 39.7-point F1 variation ($T0: 0.607 \rightarrow T3: 0.210$), larger than any architectural improvement in the literature. The T1→T2 drop (33.7 points) dwarfs the MLP-vs-GAT difference at any tier (≤ 3.6 points). This suggests that many published DEAP results reporting $>80\%$ accuracy operate in what we term the T0/T1 regime, where temporal autocorrelation inflates apparent performance.

The crossover in model rankings is informative: MLP outperforms GAT at T1 ($d = -0.83$), but GAT leads at

TABLE VI
COMPARISON WITH PUBLISHED DEAP RESULTS

Study	Protocol	Result	$\approx$ Tier
Koelstra [1]	Subj.-dep.	57.5–62.0% Acc	$\sim$ T2
High-accuracy studies [26]	Random split	90–99%	T0
Ours (T2, binary val.)	Trial-aware	64.9% F1	T2
Ours (T3, 4-class)	LOSO	21.0% F1	T3

T2 ($d = +0.74$) and T3 ($d = +0.14$). This suggests MLP over-exploits temporal autocorrelation present in Tier 1, while GAT’s spatial attention provides genuine generalization advantage visible only once leakage is removed.

B. Why Performance Degrades

Multiple factors contribute to reduced T2/T3 performance: (1) *Label noise*: DEAP self-report ratings near the per-subject median threshold are inherently ambiguous, contributing noisy labels; (2) *Autocorrelation*: adjacent windows share correlated frequency content—removing this source of apparent accuracy reveals true generalization capacity; (3) *No domain adaptation*: naive cross-subject transfer without alignment (e.g., Euclidean alignment, CORAL) is a known limitation; (4) *Task difficulty*: 4-class quadrant classification (chance=25%) is substantially harder than binary (chance=50%), amplifying all effects.

C. Controlled Stimulus Paradigm

Both datasets employ music targeting the same arousal-valence quadrants [20]. This shared paradigm enables a controlled comparison: the gap between DEAP T2 (0.311, $n=20$) and OpenBCI T2 (0.981, $n=1$) is attributable primarily to inter-subject variability rather than differences in recording hardware, feature engineering, or stimulus modality. This supports personalized EEG-BCI viability [25] while highlighting the fundamental challenge of subject-independent models.

The practical implication is clear: for consumer EEG-BCI applications (e.g., adaptive music recommendation, meditation guidance), a brief per-user calibration session could enable reliable emotion detection. However, building universal models without per-user calibration remains an open challenge requiring substantially larger training populations or domain adaptation techniques.

D. Comparison with Published Results

Table VI contextualizes our results against published DEAP benchmarks. The table confirms that the tier framework correctly categorizes published protocols: the original DEAP result falls at T2, while inflated high-accuracy studies cluster at T0.

Our T2 binary valence (0.649 F1) exceeds the original DEAP benchmark (Koelstra *et al.*: 57.5% accuracy and F1=0.563 for binary valence under subject-dependent evaluation), confirming that trial-aware evaluation produces realistic, reproducible benchmarks. Published LOSO baselines on DEAP’s arousal-valence formulation are scarce, making our T3 result (21.0% 4-class F1) a novel reference point

for cross-subject generalization difficulty. Studies reporting $>90\%$ accuracy invariably use random splits or trial-leaky CV (T0/T1). This is independently corroborated by Khan *et al.* [5], who documented that methodological errors inflate DEAP accuracy by up to 46%, and by the EEGain framework [15].

The “Watermelon” experiment [5] provides striking proof: EEG electrodes placed on a biologically inert watermelon were recorded with a 64-channel system for over one hour. The data were restructured to mirror DEAP (40 trials $\times$ 60 s) and assigned random binary labels. Under standard k -fold CV with segmentation, accuracy rose from chance to $>90\%$ with just 6 segments per trial, yet leave-one-trial-out validation correctly remained at chance ($\sim 52\%$) regardless of segmentation—demonstrating that high accuracies can be pure artifacts of temporal leakage.

Representative inflated studies include: Algarni *et al.* [27] (99.45% with 70/30 random split), Chen *et al.* [28] (100% AUC with segment-level 10-fold CV), Pichandi *et al.* [29] (97.3% with random split and globally-applied PCA), and Jin and Kim [30] (99.10% labeled “cross-subject” yet using pooled random splits where the same subject appears in both train and test). None of these studies enforce trial boundaries during cross-validation, meaning temporally adjacent and highly correlated windows appear in both training and test sets. When proper LOSO is applied, accuracy drops to 59–72% [31], aligning with our findings. Our framework reveals a 30–35 point deflation once leakage is eliminated, consistent with the 46% cumulative inflation documented by Khan *et al.* [5].

E. Limitations

This study uses 20 of 32 DEAP subjects due to computational constraints. The OpenBCI result ($n=1$) serves as proof-of-concept for personalization and requires multi-subject replication. We evaluate only two architectures; CNN-based (EEGNet [32]) and Transformer-based models may exhibit different degradation patterns. The four-tier framework does not address session effects or stimulus-order confounds. Additionally, our PLV and Coherence features are averaged per channel rather than preserved as pairwise edge weights, partially under-utilizing the graph structure that GAT could exploit.

F. Implications for Future Research

Domain adaptation: The T2 $\rightarrow$ T3 drop ($\Delta = -0.101$) suggests that subject-alignment techniques (Euclidean Alignment, CORAL, adversarial domain adaptation) [14], [25] could recover a significant portion of performance lost to inter-subject variability.

Graph structure: Our fully-connected adjacency matrix forces the GAT to learn connectivity from scratch. Neurophysiologically-informed adjacency (e.g., from resting-state coherence or anatomical distance) may improve generalization by incorporating prior knowledge about functional brain networks.

Evaluation standardization: We advocate trial-aware evaluation as the minimum standard, with explicit reporting of both T1 and T2 metrics to enable transparent quantification of leakage inflation in any proposed method.

VI. CONCLUSION

We present a four-tier evaluation framework quantifying how evaluation methodology inflates EEG emotion classification performance. Key findings: (1) protocol selection accounts for a 39.7-point macro F1 variation, dominating any architectural difference; (2) trial leakage ($\Delta = -0.337$, $d = 3.21$, $p < 0.001$) is the primary inflation source; (3) under trial-aware evaluation, GAT provides a modest but statistically significant advantage over MLP ($d = 0.74$, $p < 0.05$); (4) personalized within-subject models achieve 0.981 F1 (trial-aware), demonstrating BCI viability when inter-subject variability is eliminated; (5) binary valence classification (F1=0.649 at T2) confirms these findings generalize to the more commonly reported binary task.

We recommend trial-aware evaluation (T2) as the minimum reporting standard for EEG-BCI studies, and encourage researchers to report both T1 and T2 results to transparently quantify the role of temporal autocorrelation in their reported performance. The framework is architecture-agnostic and can be applied to any EEG classification pipeline. All code and features are publicly available at <https://github.com/mateisorodoc/Quantifying-Data-Leakage-in-EEG-Emotion-Classification> for reproducibility.

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